

# FUTURE\_IDEAS.md — Roadmap & Every Planned Feature

Scope: the ONLY place features and ideas are tracked. Priorities map to PROJECT.md → Priorities 1–4. Development is **feature-driven, not date-driven**: an item moves up only when the work above it meets the quality bar (EDUCATION.md → Quality standards).

## Where new ideas belong (routing rule)

When a new idea appears, add it to a bucket below AND, if it changes rules, update exactly one doc:

| Idea is about...                                            | Add feature here, update... |
|-------------------------------------------------------------|-----------------------------|
| A new mechanic, reward, mode, economy change                | GAME_DESIGN.md              |
| Teaching, hints, curriculum, subjects, feedback             | EDUCATION.md                |
| The kids, family, personalization, privacy of personal data | CHARACTERS.md               |
| Architecture, tooling, testing, deploy, backend, cost       | DEVELOPMENT_RULES.md        |
| Vision or priority order                                    | PROJECT.md                  |

## NOW (Priority 1: finish the math platform for Alex & Katya)

1. **Remove Leila and Emily** from all app content (mission stories, the `FRIENDS` array in the question engine, any generated text). Aryna stays (lives in Alberta). (*Content fix — do first.*)
2. **Implement the mandatory wrong-answer teaching loop** (EDUCATION.md): 3 attempts → progressive hints → step-by-step explanation with answer → 1–3 reinforcement questions → recurring-mistake detection feeding extra practice. Brain Boost is the foundation; make the flow automatic.
3. **Baseline Playwright suite** covering the minimum child-critical flows listed in DEVELOPMENT\_RULES.md → Testing.

4. **Record the live Vercel URL** in DEVELOPMENT\_RULES.md; change the parent PIN from 1234 in production.
5. **Fix XP inconsistency:** `markMissionDone` uses `xpMax × 1.15`; unify to `+50` like everywhere else.
6. **Grade/age system:** store birthdays (Dec 29 2013 / May 4 2016), auto- calculate age, current-grade field that advances each school year and is parent-adjustable; reframe "Grade 7→8 / 4→5" labels to derive from it. Summer Bridge becomes an optional mode.
7. **Move personal details into profile data** (not hardcoded strings) so they can become parent-editable fields later (CHARACTERS.md → Privacy rules).
8. **Math quality-gate audit:** check current math against the full quality checklist (EDUCATION.md) — spaced repetition, error-pattern analysis, and multiple question types are the likely gaps.
9. **Activate the achievement badges** (currently decorative): real earn conditions for First Win, Streak 7, Math Genius, Early Bird, Team Player, Perfectionist.

## NEXT (Priority 2 begins: subjects beyond math, strict order)

1. **Reading** → 2. **Grammar** → 3. **Writing** → 4. **Spelling** →
2. **Vocabulary** → 6. **French (Ontario Core French)** — each must pass the quality gate before the next starts. (Katya's existing reading/grammar/ french generators are seeds, not finished subjects.)
3. **Begin gradual React + TypeScript migration** feature-by-feature as new subjects are built (DEVELOPMENT\_RULES.md → Architecture). Zero visual/ behavioral regression; retire the dead `src/App.jsx` prototype as real React code replaces it.
4. **Text-message-style mystery challenges** for Katya (her favorite format).
5. **Bonus Missions** — optional extra sessions for highly engaged days (EDUCATION.md → Daily expectations).
6. **Russian as a third-language subject** + cultural stories (EDUCATION.md → Language policy); separate progress tracking per language.

## LATER (Priority 3: Parent Dashboard + backend)

1. **Backend migration** per DEVELOPMENT\_RULES.md → Backend path: family account, child profiles, cloud DB, cross-device sync, safe localStorage migration, offline fallback.

2. **Full Parent Dashboard:** progress by child; strengths and learning gaps; assign custom missions; create custom rewards; upload school worksheets and tests; generate personalized practice from uploads; report-card goal tracking; learning analytics; AI recommendations; manage multiple children with different learning plans; AI usage + estimated cost display.
3. **Server-side AI tutoring** (provider-abstracted; privacy rules apply): concept teaching, personalized explanations, error-pattern coaching.
4. **Remaining subjects:** Science, Social Studies, Geography, History, Financial Literacy, Coding & logical thinking, General knowledge, Study skills, Critical thinking.
5. **Ecosystem bridge** with "Just Do It and Be Done" (separate app — never merge): shared family account exchanging achievements, XP, coins, streaks, rewards. Two games, one ecosystem.

## **MAYBE (Priority 4: commercial / long horizon)**

1. **Open to other families:** convert all real names and personal details to editable profile fields with safe sample data (CHARACTERS.md → Privacy rules); onboarding, multi-family accounts, pricing.
2. **AI literacy and career exploration** modules for older grades.
3. **UI language switching** — currently explicitly out of scope (English-only UI); revisit only if the commercial product demands it.

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*Completed milestones (for context, keep short):* profile picker + parent PIN; coin/XP/Adventure Key economy; reward request→approval flow; adaptive Ontario-curriculum Daily Mission engine; Brain Boost 6-tier hint ladder + Strategy Library; Battle Arena, Money Lab, Guided Mission, Equation Escape Room, Grant Prize; weekend Double XP + weekend streak freeze; Web Audio sound system; tablet-first responsive pass; mobile input fixes (v1.1.0).